class09

Tanieng Huang

Class 9: Candy Mini-Project

2. Importing candy data

```
# Set the file name variable
candy_file <- "candy-data.csv"

# Read in the CSV; use column 1 (competitorname) as row names
candy = read.csv(candy_file, row.names=1)

# Preview the first 6 rows
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

- **Q1**. How many different candy types are in this dataset?

```r
nrow(candy)
```

```
[1] 85
```

> 85 types of candy.

- **Q2**. How many fruity candy types are in the dataset?

```r
sum(candy$fruity)
```

```
[1] 38
```

> 35 fruity candy types.

2.2 What is your favorite candy?

> `winpercent`. For a given candy this value is the percentage of people who prefer this candy over another randomly chosen candy from the dataset (what 538 term a matchup). Higher values indicate a more popular candy.

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
     winpercent
Twix   81.64291
```

- **Q3**. What is your favorite candy (other than Twix) in the dataset and what is it's `winpercent` value?

```
#my favorite candy is Reese's Peanut Butter cup
candy["Reese's Peanut Butter cup", ]$winpercent
```

```
[1] 84.18029
```

```
# Answer: 84.18029
```

- **Q4**. What is the `winpercent` value for "Kit Kat"?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

```
# Answer: 76.7686
```

- **Q5**. What is the `winpercent` value for "Tootsie Roll Snack Bars"?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

```
# Answer: 49.6535
```

```
library("skimr")
```

```
Warning: package 'skimr' was built under R version 4.5.3
```

```
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12

Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

- **Q6**. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

  Yes, the winpercent column is on a 0–100 scale, while all other columns are on a 0–1 scale. From the `skim()` output which winpercent has a mean around ~50 and max around ~84. This scale difference is why we need scale=TRUE when doing PCA later.

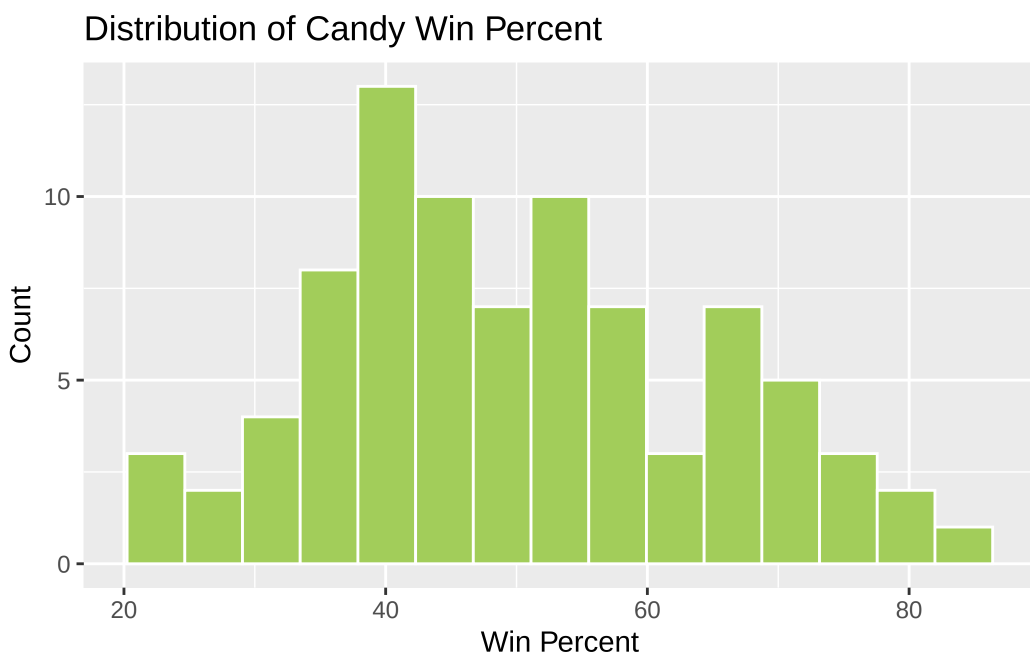
- **Q7**. What do you think a zero and one represent for the `candy$chocolate` column?

  A 0 means the candy does not contain chocolate; a 1 means it does contain chocolate. These are logical/binary indicator variables (FALSE/TRUE encoded as integers).

3 Exploratory analysis

- **Q8**. Plot a histogram of `winpercent` values using both base R an **ggplot2**.

```r
library(ggplot2)
ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins = 15, fill = "darkolivegreen3", col = "white") +
  labs(title = "Distribution of Candy Win Percent",
       x = "Win Percent", y = "Count")
```



- **Q9**. Is the distribution of `winpercent` values symmetrical?

  No. The distribution is slightly skewed to the right, meaning there are a few candies with notably high win percentages that pull the tail to the right, while the bulk of candies cluster around 40–55%

- **Q10**. Is the center of the distribution above or below 50%?

```r
median(candy$winpercent)
```

```
[1] 47.82975
```

```r
mean(candy$winpercent)
```

```
[1] 50.31676
```

The center is below 50% (~47.8%).

- **Q11**. On average is chocolate candy higher or lower ranked than fruit candy?

```
# Mean winpercent for chocolate candies
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```
# Mean winpercent for fruity candies
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

Chocolate candy is ranked higher on average (~60.9%) than fruity candy (~44.1%).

- **Q12**. Is this difference statistically significant?

```
t.test(candy$winpercent[as.logical(candy$chocolate)],
       candy$winpercent[as.logical(candy$fruity)])
```

```

	Welch Two Sample t-test

data:  candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$f
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y 
 60.92153  44.11974 
```

Yes. The t-test yields a p-value of ~2.87e-08, which is below 0.05. This means the difference in average win percentage between chocolate and fruity candies is statistically significant.

4 Overall Candy Rankings

- **Q13**. What are the five least liked candy types in this set?

```r
# Using base R
head(candy[order(candy$winpercent),], n=5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

The five least popular candies are: Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

- **Q14**. What are the top 5 all time favorite candy types out of this set?

```r
tail(candy[order(candy$winpercent),], n=5)
```

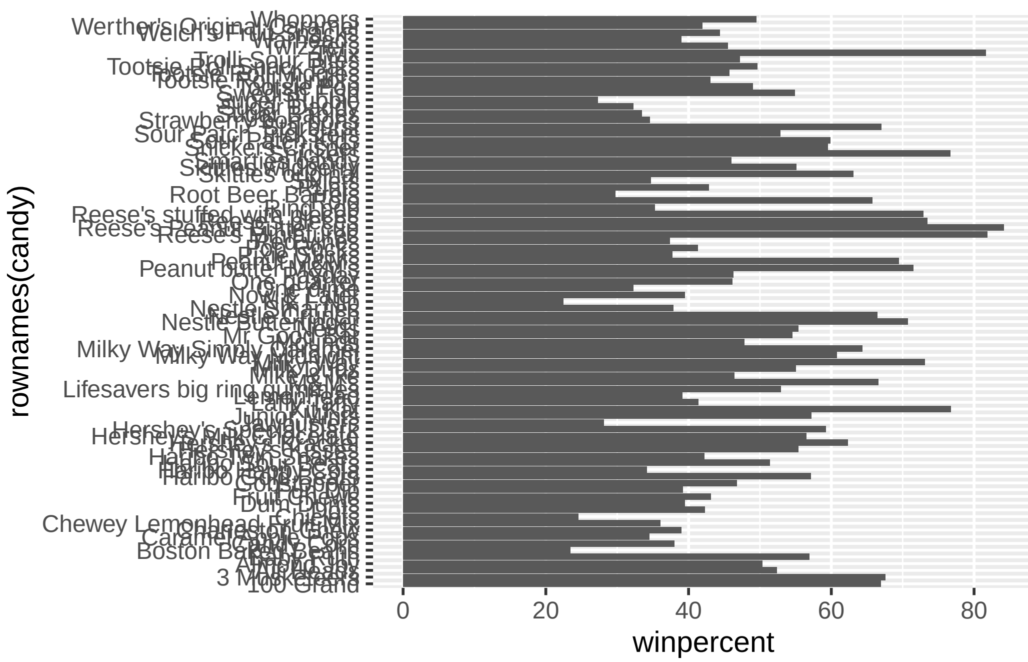
```
                          chocolate fruity caramel peanutyalmondy nougat
Snickers                          1      0       1              1      1
Kit Kat                           1      0       0              0      0
Twix                              1      0       1              0      0
Reese's Miniatures                1      0       0              1      0
Reese's Peanut Butter cup         1      0       0              1      0
                          crispedricewafer hard bar pluribus sugarpercent
Snickers                                 0    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Twix                                     1    0   1        0        0.546
Reese's Miniatures                       0    0   0        0        0.034
Reese's Peanut Butter cup                0    0   0        0        0.720
```

```
                          pricepercent winpercent
Snickers                         0.651   76.67378
Kit Kat                          0.511   76.76860
Twix                             0.906   81.64291
Reese's Miniatures               0.279   81.86626
Reese's Peanut Butter cup        0.651   84.18029
```

The top 5 favorites are: Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

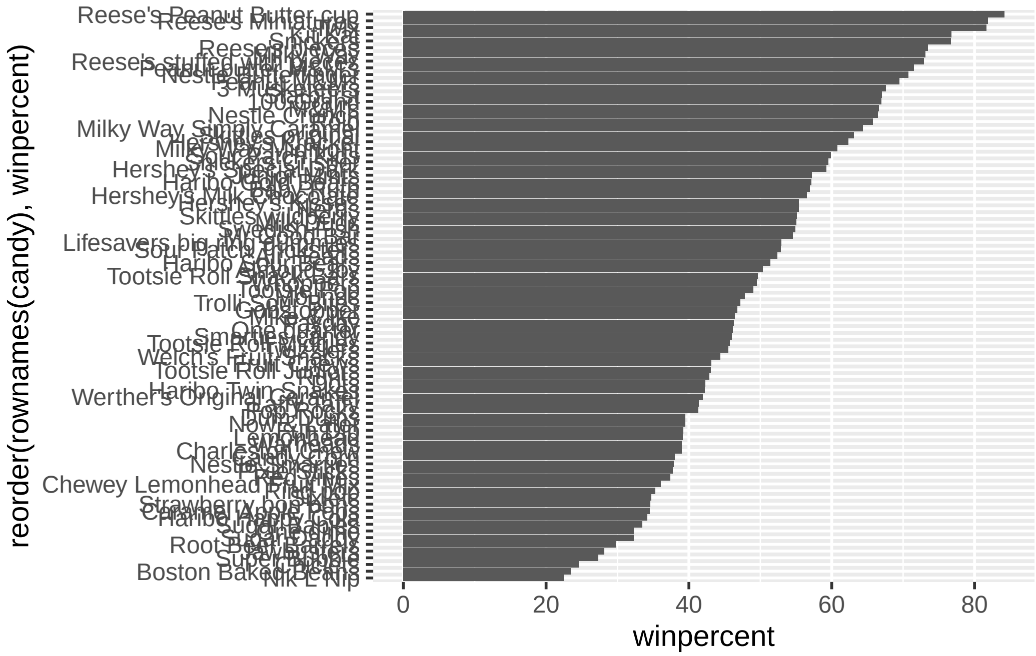
- **Q15**. Make a first barplot of candy ranking based on `winpercent` values.

```r
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



- **Q16**. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

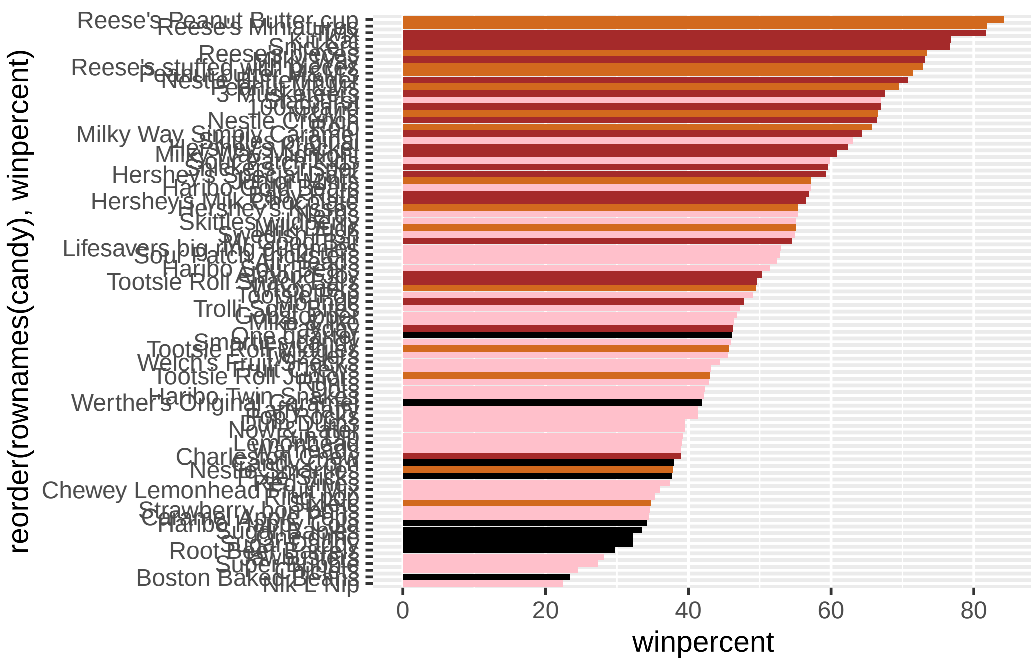
```r
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```



4.0.1 Time to add some useful color

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```



- **Q17**. What is the worst ranked chocolate candy?

  The lowest chocolate-colored bar is Sixlets.

- **Q18**. What is the best ranked fruity candy?

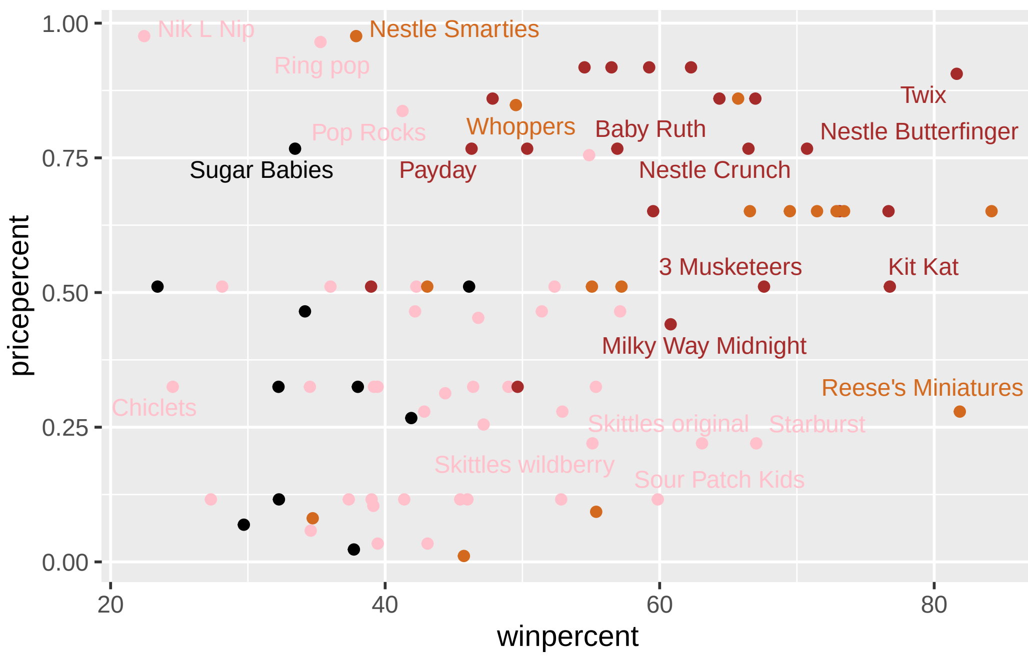
  The highest fruity bar is Starburst.

5 Taking a look at pricepercent

```
library(ggrepel)
```

```
Warning: package 'ggrepel' was built under R version 4.5.3
```

```
# Plot winpercent vs pricepercent with labeled points
ggplot(candy) +
  aes(winpercent, pricepercent, label = rownames(candy)) +
  geom_point(col = my_cols) +
  geom_text_repel(col = my_cols, size = 3.3, max.overlaps = 5)
```



- **Q19**. Which candy type is the highest ranked in winpercent for the least money - i.e. offers the most bang for your buck?

  Reese's Miniatures has the highest winpercent values but a lowest pricepercent.

- **Q20**. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
# Find the 5 most expensive
ord <- order(candy$pricepercent, decreasing = TRUE)
head(candy[ord, c("pricepercent", "winpercent")], n = 5)
```

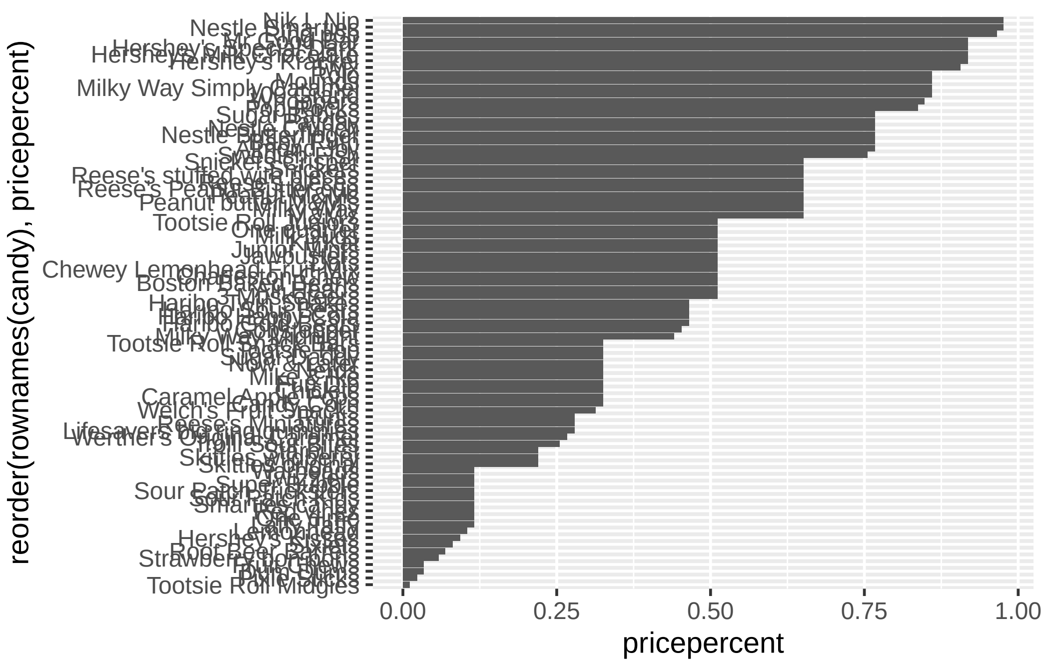
```
                         pricepercent winpercent
Nik L Nip                       0.976   22.44534
Nestle Smarties                 0.976   37.88719
Ring pop                        0.965   35.29076
Hershey's Krackel               0.918   62.28448
Hershey's Milk Chocolate        0.918   56.49050
```

The 5 most expensive candies are: Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate. The least popular is Nik L Nip.

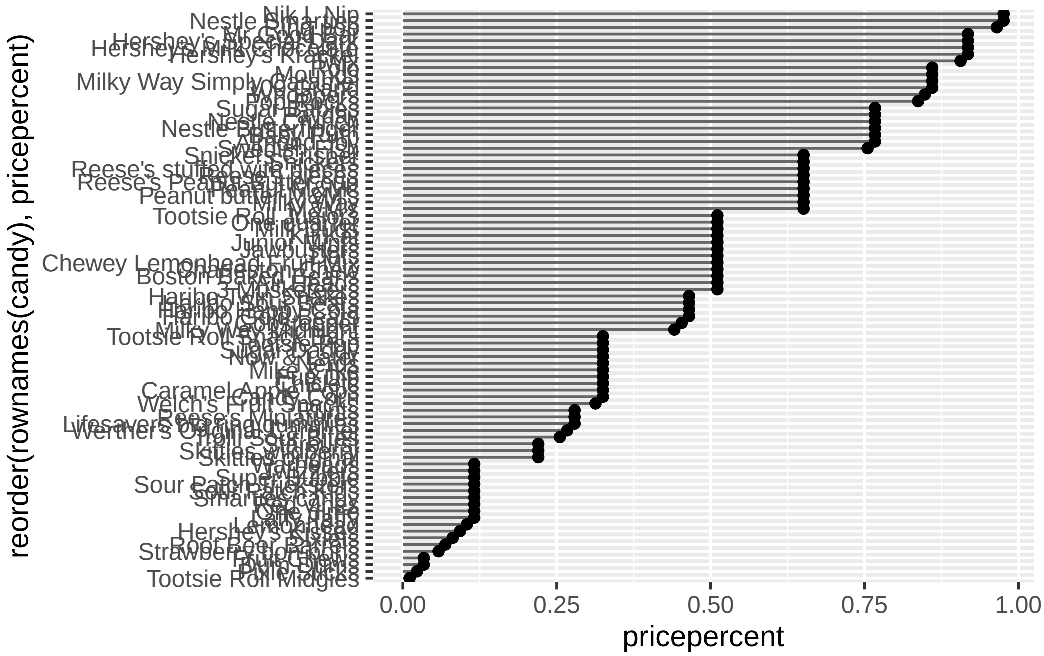
5.1 Optional

- **Q21**. Make a barplot again with `geom_col()` this time using `pricepercent` and then improve this step by step, first ordering the x-axis by value and finally making a so called "dot chat" or "lollipop" chart by swapping `geom_col()` for `geom_point()` + `geom_segment()`.

```
# Barplot ordered by pricepercent
ggplot(candy) +
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +
  geom_col()
```




```
# Lollipop chart version
ggplot(candy) +
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +
  geom_segment(aes(yend = reorder(rownames(candy), pricepercent),
                   xend = 0), col = "gray40") +
  geom_point()
```



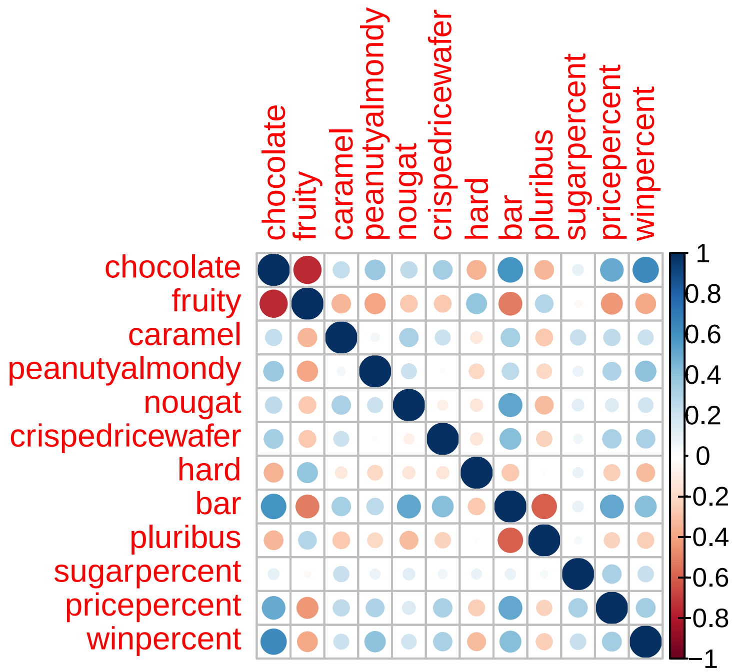
6 Exploring the correlation structure

```r
library(corrplot)
```

```
Warning: package 'corrplot' was built under R version 4.5.3
```

```
corrplot 0.95 loaded
```

```r
cij <- cor(candy)
corrplot(cij)
```



- **Q22**. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

  Chocolate and fruity are the most strongly anti-correlated variables.

- **Q23**. Similarly, what two variables are most positively correlated?

  Chocolate and winpercent also with chocolate and bar are strongly positively correlated.

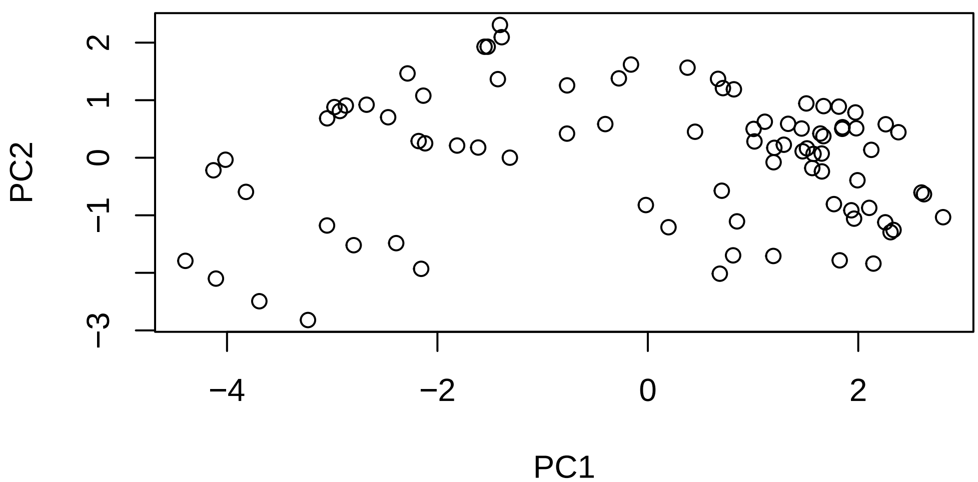
7 Principal Component Analysis

```
# Run PCA with scaling (because winpercent is on a different scale)
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

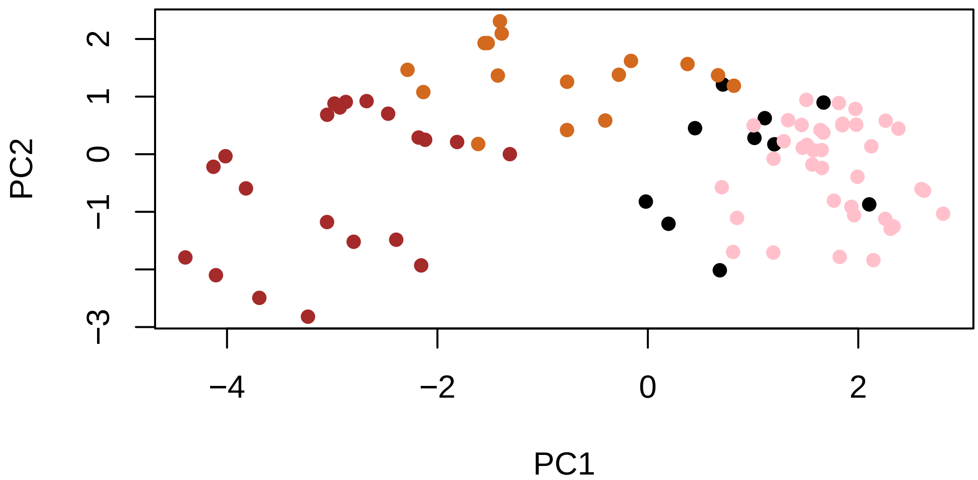
```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
```

```
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

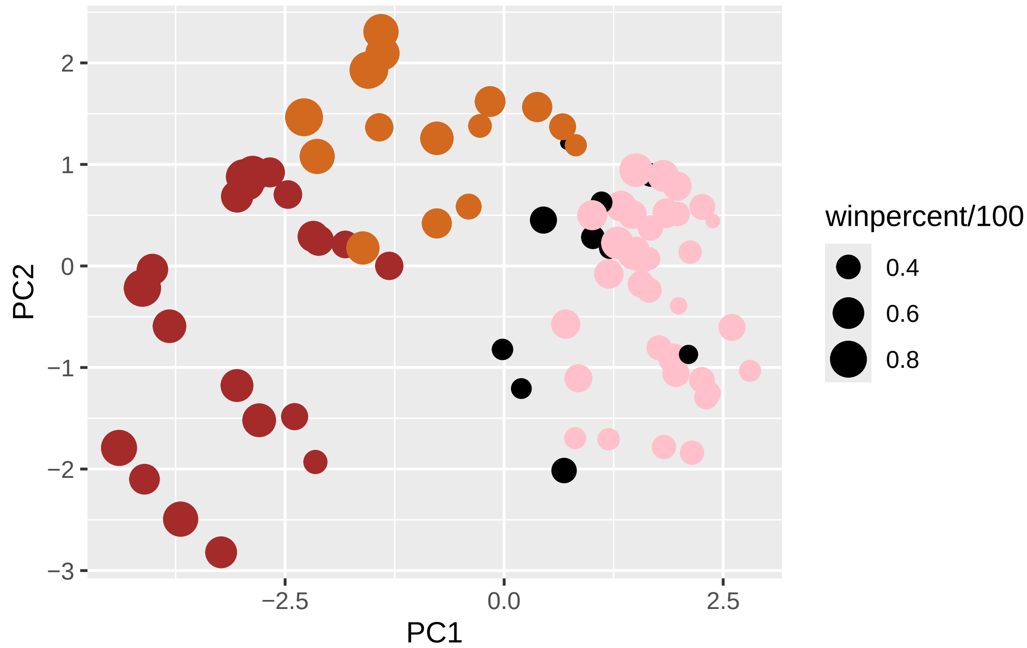
```r
# Base R plot of PC1 vs PC2
plot(pca$x[, 1:2])
```



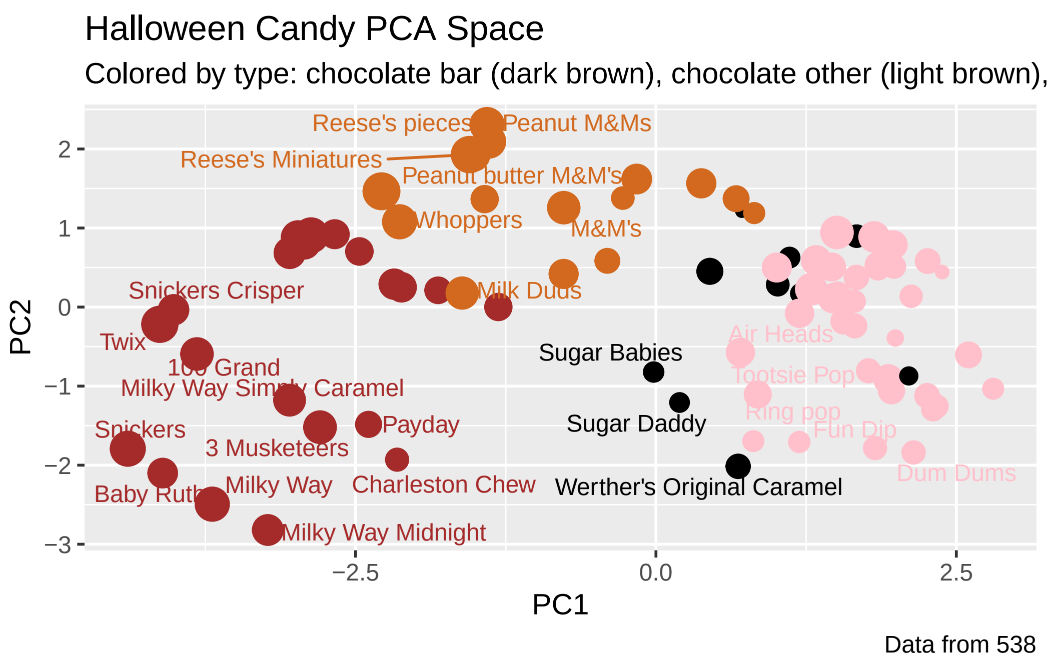

```r
# With color
plot(pca$x[, 1:2], col = my_cols, pch = 16)
```




```
# Combine candy data with PCA scores
my_data <- cbind(candy, pca$x[, 1:3])

# Create ggplot
p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,
      size = winpercent/100,
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = my_cols)
p
```




```r
library(ggrepel)

p + geom_text_repel(size = 3.3, col = my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title = "Halloween Candy PCA Space",
       subtitle = "Colored by type: chocolate bar (dark brown), chocolate other (light brown)
       caption = "Data from 538")
```

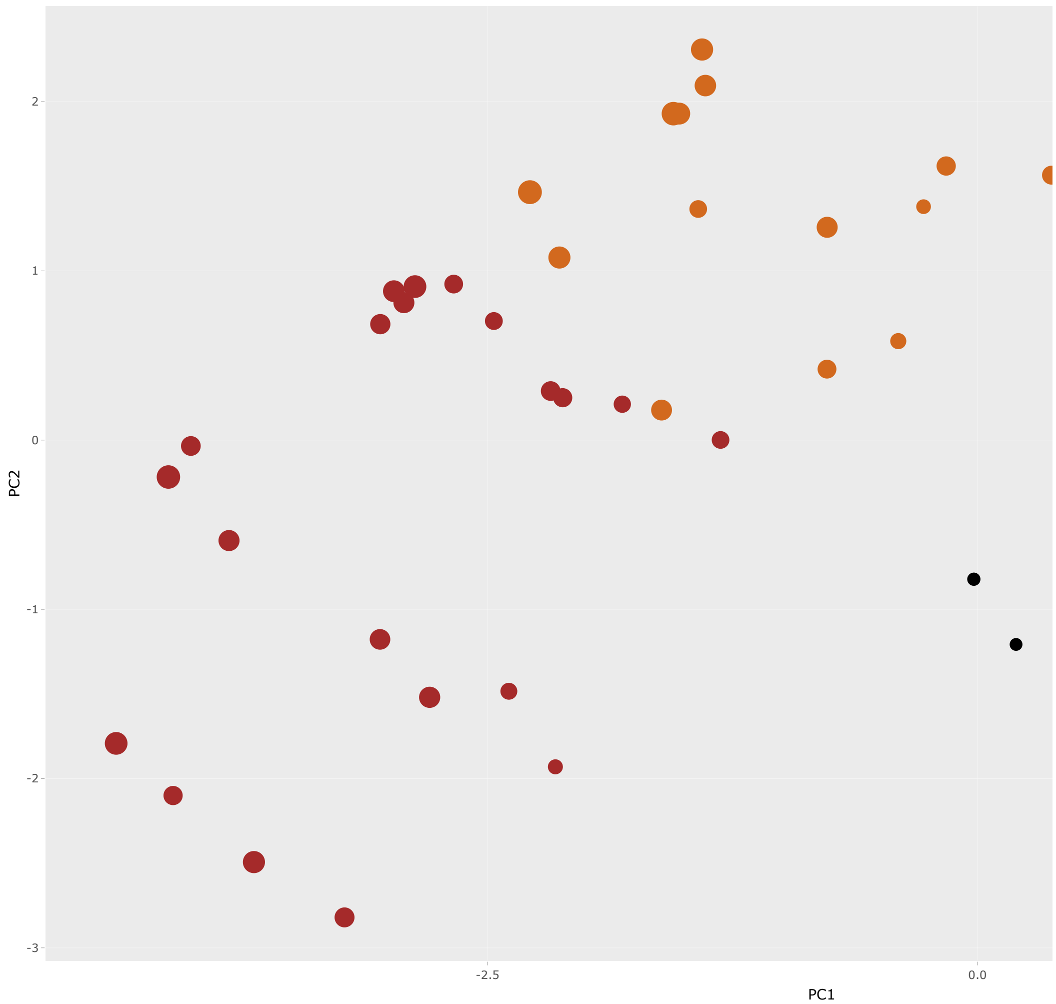


interactive plot

```
library(plotly)
```

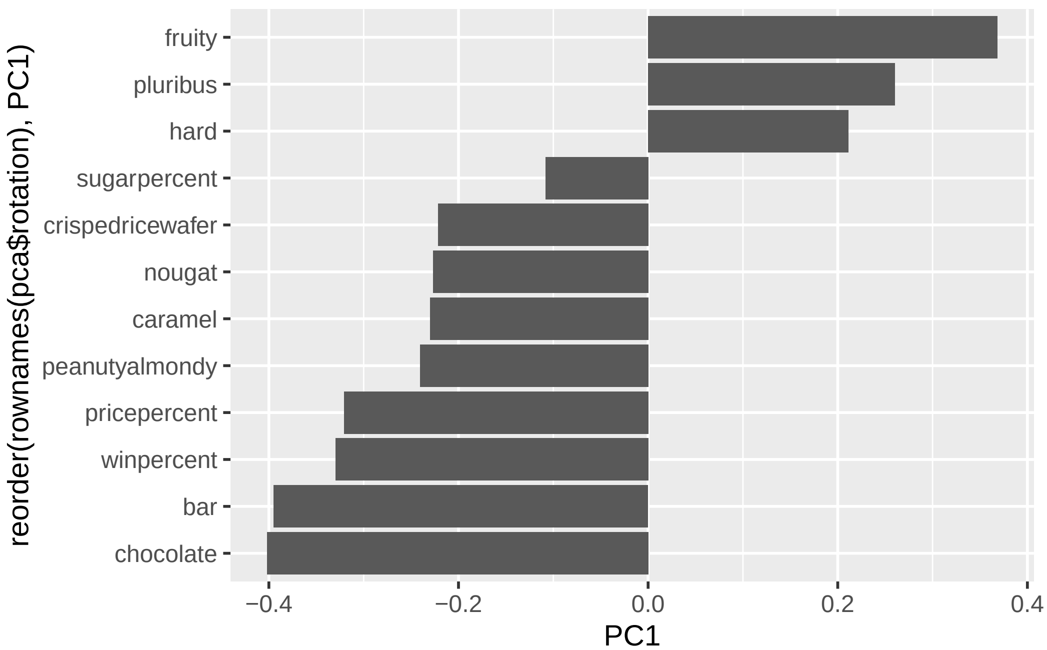
```
Warning: package 'plotly' was built under R version 4.5.3


Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

    last_plot

The following object is masked from 'package:stats':

    filter

The following object is masked from 'package:graphics':

    layout
```

```r
ggplotly(p)
```


2
1
0
-1
-2
-3
PC2
-2.5
0.0
PC1

Loading plot:

```
# Loadings plot: shows how each original variable contributes to PC1
ggplot(pca$rotation) +
  aes(PC1, reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



- **Q24**. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

  The variables with the strongest positive loadings on PC1 are fruity, pluribus, and hard. This means candies on the right side of the PCA plot tend to be fruity, come in bags/boxes of multiples, and are hard candies. This mirrors the correlation matrix: chocolate and fruity are anti-correlated, chocolate is correlated with bar, caramel, and etc. PCA captures this same structure but reduces it to a single interpretable axis.

- **Q25**. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Based on the combined analyses, winning candies tend to be:

Chocolate-based, whichthe bar plots showed chocolate candies dominate the top rankings, the correlation matrix confirmed chocolate is positively correlated with winpercent, and in PCA, the high-winpercent candies cluster on the chocolate (left/negative PC1) side.

Bar format: the top candies are bars, and bar is positively correlated with winpercent.

Contain peanuts, almonds or caramel: peanuty, almondy and caramel are associated with higher rankings.

- **Q26**. Are popular candies more expensive? In other words: is price significantly different between “winners” and “losers”? List both average values and a P-value along with your answer.

```r
losers <- candy[which(candy$winpercent < 50), ]
winners <- candy[which(candy$winpercent >= 50), ]

mean(winners$pricepercent)
```

```
[1] 0.580359
```

```r
mean(losers$pricepercent)
```

```
[1] 0.3743696
```

```r
t.test(winners$pricepercent, losers$pricepercent)
```

```

	Welch Two Sample t-test

data:  winners$pricepercent and losers$pricepercent
t = 3.5653, df = 82.798, p-value = 0.0006068
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.09107157 0.32090727
sample estimates:
mean of x mean of y
0.5803590 0.3743696
```

Yes, winners are significantly more expensive on average. Winners have a mean pricepercent of ~0.596 vs. ~0.419 for losers. The t-test p-value (~0.005) is below 0.05, so this difference is statistically significant. This could mean that more desirable candies naturally cost more to produce, or that price itself is associated with perceived quality.

- **Q27**. Are candies with more sugar more likely to be popular? What is your interpretation of the means and P-value in this case?

```r
mean(winners$sugarpercent)
```

```
[1] 0.5343077
```

```r
mean(losers$sugarpercent)
```

```
[1] 0.4314565
```

```r
t.test(winners$sugarpercent, losers$sugarpercent)
```

```

	Welch Two Sample t-test

data:  winners$sugarpercent and losers$sugarpercent
t = 1.6958, df = 81.799, p-value = 0.09373
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.01780701  0.22350935
sample estimates:
mean of x mean of y 
0.5343077 0.4314565 
```

```r
# p-value ~ 0.046
```

Winners have a higher mean sugar percentile (~0.53) than losers (~0.42). The p-value (~0.046) is lower than 0.05 threshold, indicating a statistical significant difference. Yet, this significance is marginal, meaning the effect is relatively weak. While higher sugar content is associated with higher winpercent, the small difference in means (~0.11) suggests that sugar alone does not strongly determine candy popularity.